极限荷载作业及答案

10－1 图 10－17 所示四种单跨梁，其材料、截面均相同，则极限荷载值最大的是（　　）。（浙江大学 2012）

A. 图（a）　　B. 图（b）　　C. 图（c）　　D. 图（d）

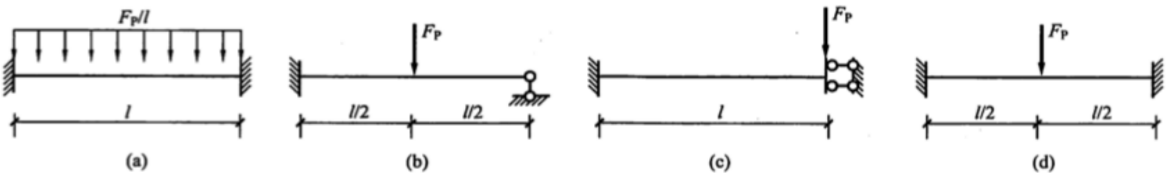


图 10－17

10－2 图 10－17 所示四种单跨梁，其材料、截面均相同，则极限荷载值最小的是（　　）。（浙江大学 2012）

A. 图（a）　　B. 图（b）　　C. 图（c）　　D. 图（d）

10－3 图 10－18 所示梁的极限弯矩为 M_u，则其极限荷载 F_{Pu}＝______。（浙江大学 2009）

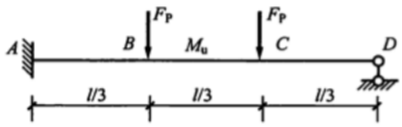


图 10－18

10－4 如图 10－19 所示矩形截面连续梁，材料屈服极限 σ_s，AB 跨截面宽 b 高 h，其极限弯矩 M_u＝______，设 BC 跨的极限弯矩为 $1.5M_u$，其极限荷载 F_{Pu}＝______（北京工业大学 2009）

10－5 求图 10－20 所示结构的极限荷载 q_u。（南京工业大学 2011）

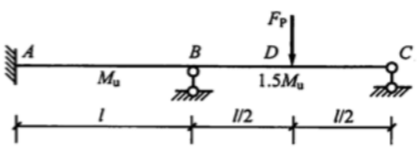


图 10－19

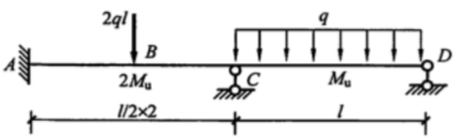


图 10－20

10－6 计算图 10－21 所示连续梁的极限荷载 F_{Pu}。（青岛理工大学 2011）

10－7 试求图 10－22 所示多跨梁的极限荷载 q_u。（青岛理工大学 2008）

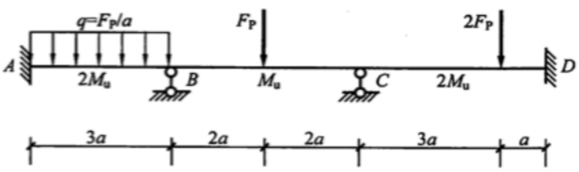


图 10－21

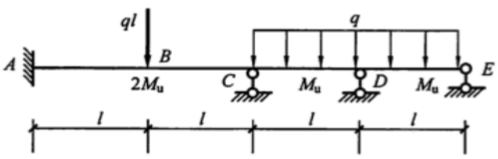


图 10－22

练习题答案

10－1　A。**提示：**$F_{Pu}=16\dfrac{M_u}{l}$。

10-2　C。**提示：**$F_{Pu}=2\frac{M_u}{l}$，破坏时两端截面出现塑性铰。图 10-17（b）$F_{Pu}=6\frac{M_u}{l}$，图 10-17（d）$F_{Pu}=8\frac{M_u}{l}$。

10-3　$4\frac{M_u}{l}$。**提示：**相应的破坏机构为截面 A、C 出现塑性铰。

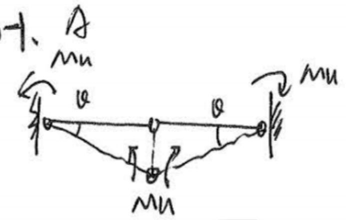
10-4　$\frac{bh^2}{4}\sigma_s$；$8\frac{M_u}{l}$。**提示：**相应的破坏机构为截面 B、D 出现塑性铰。

10-5　$7\frac{M_u}{l^2}$。**提示：**相应的破坏机构为截面 A、B、C 出现塑性铰。

10-6　$F_{Pu}=2\frac{M_u}{a}$。**提示：**BC 跨形成机构。

10-7　$7\frac{M_u}{l^2}$。**提示：**相应的破坏机构为截面 A、B、C 出现塑性铰。

10-1、D

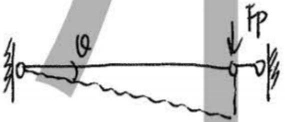


$$M_u\cdot\theta+M_u\cdot\theta+M_u\cdot2\theta=\frac{1}{2}\times F_P\times 1\times\theta\times\frac{l}{2}$$

$$\therefore\ 4M_u=\frac{F_Pl}{4}\quad\therefore\ F_P=\frac{16M_u}{l}$$

$$\theta M_u+M_u\cdot2\theta=F_P\cdot\theta\cdot\frac{l}{2}$$

$$\therefore\ 3M_u\theta=\frac{F_Pl\theta}{2}\quad\therefore\ F_P=\frac{6M_u}{l}$$



$$M_u\cdot\theta+M_u\cdot\theta=F_P\cdot\theta\cdot l\quad\therefore\ F_P=\frac{2M_u}{l}$$

$$M_u\cdot\theta+M_u\cdot\theta+M_u\cdot2\theta=F_P\cdot\frac{l}{2}\theta$$

$$\therefore\ 4M_u\cdot\theta=\frac{F_Pl\theta}{2}\quad\therefore\ F_P=\frac{8M_u}{l}$$

10-2、C

10-3、

$$M_u\cdot\theta+M_u\cdot3\theta=F_P\cdot\theta\cdot\frac{l}{3}+F_P\cdot2\theta\cdot\frac{l}{3}$$

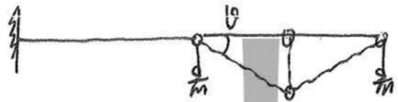
$$\therefore\ 4M_u\cdot\theta=F_P\cdot\theta\cdot l$$

$$\therefore\ F_P=\frac{4M_u}{l}$$

10-4.

$\sigma_s \times b \times \frac{h}{2} \times \frac{h}{2} = M_u$

$\therefore M_u = \frac{bh^2}{4}\sigma_s$

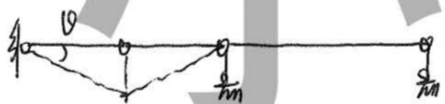


$M_u\cdot\theta + 1.5M_u\cdot 2\theta = F_P \times \frac{l}{2}\times\theta$

$\therefore 4M_u\cdot\theta = \frac{F_P l}{2}\times\theta$

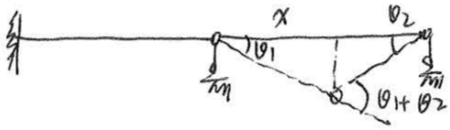
$\therefore F_{Pu} = \frac{8M_u}{l}$

10-5.



$2M_u\cdot\theta + 2M_u\cdot 2\theta + M_u\cdot\theta = 2ql\times\frac{l}{2}\times\theta$

$\therefore 7M_u\cdot\theta = ql^2\cdot\theta$ $\therefore q_u = \frac{7M_u}{l^2}$



$M_u\cdot\theta_1 + M_u\cdot(\theta_1+\theta_2) = \frac{1}{2}\times q_u \times l \times \theta_1 x$ $\frac{y}{x} = \theta_1$

$M_u(\theta_1+\theta_1+\theta_2) = \frac{lx\theta_1}{2}q_u$ $\frac{y}{l-x} = \theta_2$

$\therefore q_u = \frac{2M_u(\theta_1+\theta_2+\theta_1)}{lx\cdot\theta_1}$ $\therefore \frac{l-x}{x} = \frac{\theta_1}{\theta_2}$

$\therefore q_u = \frac{2M_u}{l}\cdot\frac{2l-x}{(l-x)\cdot x}$ $\therefore \theta_2 = \frac{x\cdot\theta_1}{l-x}$

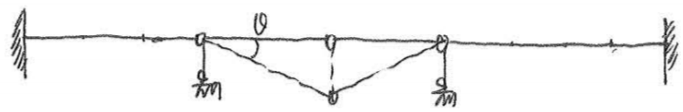
$\frac{dq_u}{dx} = 0$ $\therefore x = (2-\sqrt{2})l$

$\therefore q_u = \frac{11.66M_u}{l^2}$

$\therefore q_u = \frac{7M_u}{l^2}$

10-6、

①



$$M_u\cdot\theta + M_u\cdot\theta + M_u\cdot 2\theta = F_P\times\theta\times 2a$$

$$\therefore 4M_u\theta = 2a\times F_P\times\theta$$

$$\therefore F_P = \frac{2M_u}{a}$$

②

$$M_u\cdot\theta + 2M_u\cdot 4\theta + 2M_u\cdot 3\theta = 2F_P\cdot 3a\cdot\theta$$

$$15M_u\cdot\theta = 6F_P\cdot a\theta$$

$$\therefore F_P = \frac{15M_u}{6a}$$

③

$$2M_u\cdot\theta_1 + M_u\cdot\theta_2 + 2M_u(\theta_1+\theta_2) = \frac{1}{2}\times 3a\times x\theta_1\times\frac{F_P}{a}$$

$$M_u(2\theta_1+\theta_2+2\theta_1+2\theta_2) = \frac{3x\theta_1}{2}F_P$$

$$M_u(4\theta_1+3\theta_2) = \frac{3x\theta_1}{2}F_P \qquad \therefore F_P = \frac{2M_u(4\theta_1+3\theta_2)}{3x\theta_1}$$

$$\theta_1 = \frac{y}{x} \qquad \therefore \frac{\theta_1}{\theta_2} = \frac{3a-x}{x} \qquad \therefore \theta_2 = \frac{\theta_1 x}{3a-x}$$

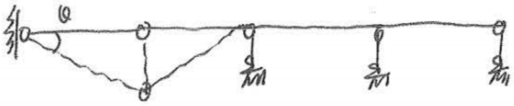
$$\theta_2 = \frac{y}{3a-x} \qquad \therefore F_P = \frac{2M_u}{3}\cdot\frac{x-12a}{x^2-3ax} \qquad \frac{\partial F_P}{\partial x} = \frac{x^2+24ax-36a^2}{x^2-3ax}\cdot\frac{2M_u}{3} = 0$$

$$\therefore x = 1.61a \qquad \therefore F_P = \frac{4.65M_u}{a}$$

$$\therefore F_P = \frac{2M_u}{a}$$

10-7.

①



$2M_u\cdot\theta + 2M_u\cdot 2\theta + M_u\cdot\theta = ql\times\theta\times l$

∴ $7M_u\cdot\theta = ql^2\cdot\theta$ ∴ $q_u = \frac{7M_u}{l^2}$

②

$M_u\theta_1 + M_u\theta_2 + M_u(\theta_1+\theta_2) = \frac{1}{2}\times l\times\theta_1 x\cdot q$

$2M_u(\theta_1+\theta_2) = \frac{l\theta_1}{2}x\cdot q$

∴ $q = \frac{4M_u(\theta_1+\theta_2)}{l\theta_1 x}$

$\frac{dq}{dx}=0$ ∴ $x=\frac{l}{2}$

∴ $q_u = \frac{16M_u}{l^2}$

$\theta_1 = \frac{y}{x}$

$\theta_2 = \frac{y}{l-x}$

∴ $\frac{\theta_1}{\theta_2} = \frac{l-x}{x}$

∴ $\theta_2 = \frac{\theta_1 x}{l-x}$

③

$M_u\theta_1 + M_u(\theta_1+\theta_2) = \frac{l\theta_1}{2}xq$

$M_u(2\theta_1+\theta_2) = \frac{l\theta_1}{2}xq$

∴ $q = \frac{2M_u(2\theta_1+\theta_2)}{l\theta_1 x}$ $\frac{dq}{dx}=0$ ∴ $x=(2-\sqrt{2})l$

∴ $q = \frac{11.66M_u}{l^2}$